Review for the Second Midterm Exam

1. Second Midterm Exam
   - on Friday 13 November from 1:00pm till 1:50pm
   - skipping policy
2. Calculus, Plotting, and Solving
   - functions, differentiation, integration, series
   - plotting and solving

MCS 320 Lecture 34
Introduction to Symbolic Computation
Jan Verschelde, 11 November 2026

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on Friday 13 November from 1:00pm till 1:50pm

- You must sign up before noon on Thursday 12 November by sending an email that you commit to taking the exam.
- Without signing up, you will not receive the questions.
- You must solve the questions on the exam *individually*.
- The exam will have multiple versions of the questions.
- The answers must be uploaded into gradescope
  - in one single Jupyter notebook,
  - before 1:50pm on Friday 13 November 2026.

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Skipping Policy

- You skip the exam automatically if you do not do anything.
- If you skip a midterm exam, then the 100 points are transferred to the final exam, which will count for 100 points more.
- If you have seen the questions, but do not submit your answers, then you will receive 0/100.
- If you have taken the exam, then your score counts.

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Functions, Differentiation, Integration, Series

Lectures 18 to 24 cover the third part.

- Know the difference between SageMath and Python functions.
- Memoization makes recursion efficient.
- We defined functions to compute expressions.
- The slope of the tangent line at a point on a curve is computed via implicit differentiation or via a linear Taylor series approximation.
- If a symbolic antiderivative does not exist, then we can numerically evaluate a definite integral, except for singularities in the interval.
- A symbolic-numeric factorization works over complex intervals.
- Lagrange multipliers solve constrained optimization problems.

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Plotting and Solving

Lectures 25 to 31 cover the fourth part.

- In two dimensions, we have four different types of plots: explicit, implicit, parametric, and polar plots.
- In three dimensions, we distinguish between surfaces and curves.
- An animation is a list of plots, called frames.
- We solve polynomial equations by triangular forms.
- In linear algebra, we compute matrix factorizations.
- Differential equations occur in mathematical modeling.
- We ended with polyhedral and unconstrained optimization.